

FIGURE 3: Synthetic Benchmark Case Families, Partition Geometry, and Truth Taxonomy

A Case Family Taxonomy (20 Families × 3 Partitions = 380 Cases)

Total Benchmark: 380 Cases | 80 Development (4/fam) | 240 Held-Out (12/fam) | 60 Challenge (3/fam)

F01 Noiseless M0 4 Dev 12 Held 3 Ch	F02 Mod Noise M0 4 Dev 12 Held 3 Ch	F03 Strong Noise 4 Dev 12 Held 3 Ch	F04 Missing Energy 4 Dev 12 Held 3 Ch	F05 Boundary Scale 4 Dev 12 Held 3 Ch
F06 Null Scalar (M1) 4 Dev 12 Held 3 Ch	F07 Mass-Only g 4 Dev 12 Held 3 Ch	F08 Linear Desc 4 Dev 12 Held 3 Ch	F09 Saturating Desc 4 Dev 12 Held 3 Ch	F10 Interaction Law 4 Dev 12 Held 3 Ch
F11 Irrelevant Dist 4 Dev 12 Held 3 Ch	F12 Correlated Dist 4 Dev 12 Held 3 Ch	F13 M1 Violation 4 Dev 12 Held 3 Ch	F14 M2 High Floor 4 Dev 12 Held 3 Ch	F15 M3 Low Ceiling 4 Dev 12 Held 3 Ch
F16 M1+M2+M3 Comb 4 Dev 12 Held 3 Ch	F17 Equivalent Form 4 Dev 12 Held 3 Ch	F18 Exponential Law 4 Dev 12 Held 3 Ch	F19 Null Targets (A-C) 4 Dev 12 Held 3 Ch	F20 Adversarial (A-C) 4 Dev 12 Held 3 Ch

B Within-Case Geometry (180 Compounds in 30 Scaffold Groups)

Disjoint Scaffold Group Partitioning (6 Compounds / Scaffold)

Training Partition (20 Scaffolds / 120 Cmpds)

Scaffolds S001 – S020

- Shared profile $\Phi(u)$ fit
- Symbolic search $\hat{g} = f(\mathbf{x})$
- Centering & variance fitting
- 6 collision energies per cmpd

Validation Set (5 Scaffolds)

S021 – S025 (30 Cmpds)
Null fit & Pareto selection

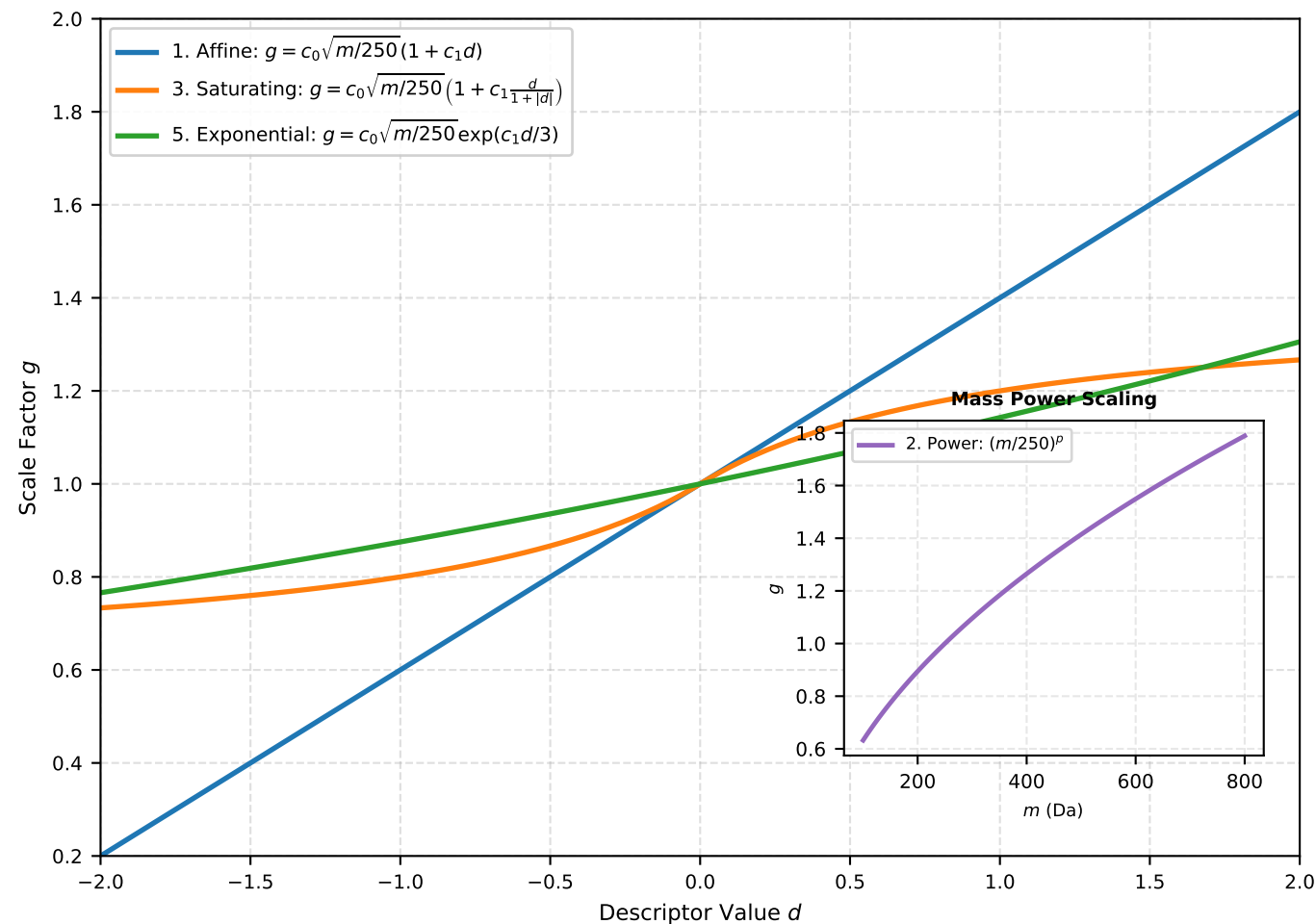
Test Set (5 Scaffolds)

S026 – S030 (30 Cmpds)
Adequacy & GBDT ceiling

Five Primitive Benchmark Covariates & Energy Grid

- mass**: Precursor mass $m \in [100, 800]$ Da (log-uniform scaffold distribution)
 - descriptor**: Active continuous molecular descriptor $d \in [-2.5, +2.5]$
 - descriptor2**: Secondary descriptor $d_2 \in [-2.5, +2.5]$ (for interaction family F10)
 - distractor**: Independent orthogonal Gaussian noise $z \sim \mathcal{N}(0, 1)$
 - correlated_distractor**: Collinear nuisance variable $z_{\text{corr}} = 0.85d + 0.15\epsilon$
- Energy Grid**: Collision energy $E \in \{15, 30, 45, 60, 75, 90\}$ NCE (6 points)

C Five Truth-Family Taxonomy Functional Forms



D Adequacy Violation Models (M0 vs M1, M2, M3 Standalone & F16 Combined)

